

Preliminary figures request:

Thermal dielectron production and temperature measurement in 0+0 collisions at $\sqrt{s_{NN}} = 200$ GeV

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Previous slides:

- <https://drupal.star.bnl.gov/STAR/meetings/STAR-Collaboration-Meeting-March-2025/LFSUPC-PWG-Parallel-Session-I/Raw-signal-dielectron-p>
- <https://drupal.star.bnl.gov/STAR/event/2026/01/26/LFSUPC-PWG-meeting/Update-dielectron-measurement-00-collisions-200-GeV>
- <https://drupal.star.bnl.gov/STAR/meetings/STAR-Collaboration-Meeting-March-2026/LFSUPC-PWG-parallel-session-III/Update-thermal-dielec>



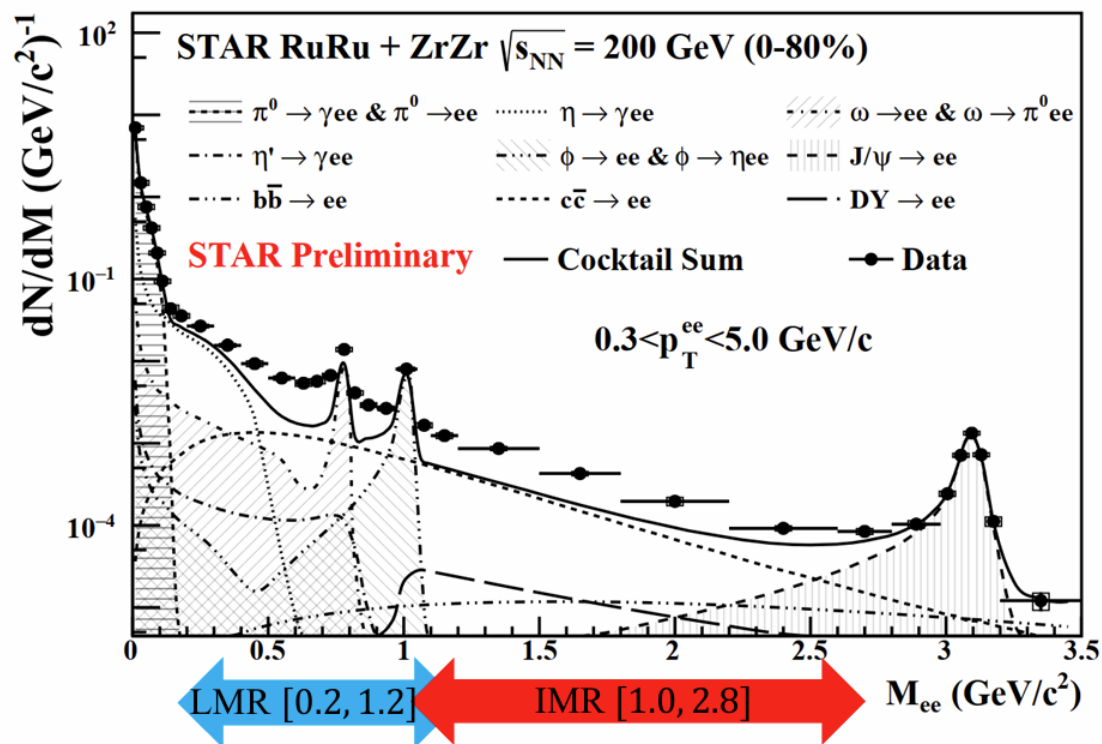


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Motivation

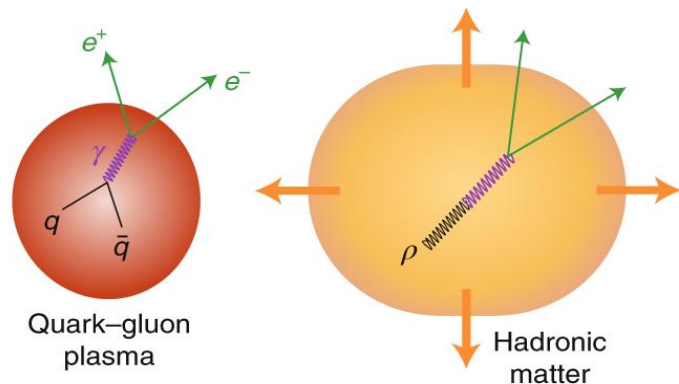


- Dileptons are ideal probes to study the evolution of heavy-ion collisions:
 - Emitted **from early to final stages**.
 - **No strong interaction** with medium.
 - **No blue-shift effect**.
- High Mass Range:
 - Drell-Yan, Heavy quarkonia.
- Intermediate Mass Range:
 - Semi-leptonic decay of heavy flavor quarks.
 - **QGP thermal radiation**.
- Low Mass Range:
 - Hadron two-body decay & Dalitz decay.
 - **In-medium vector mesons**.

Motivation

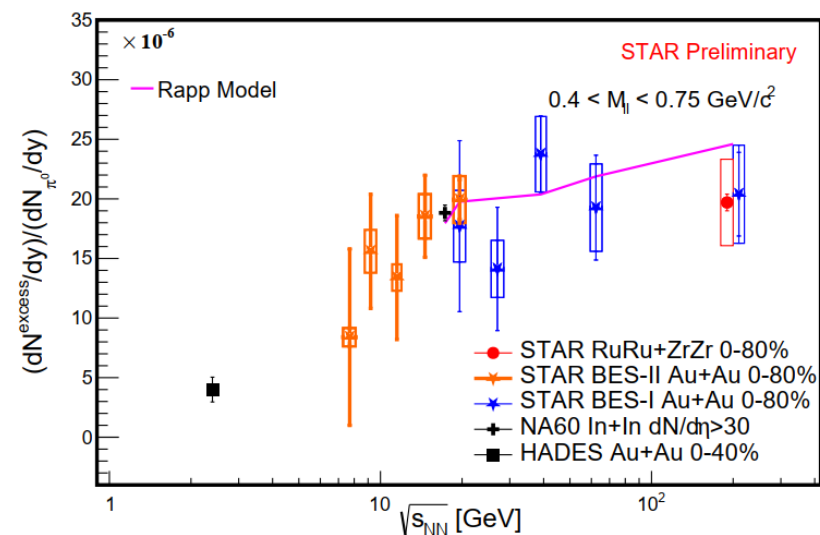
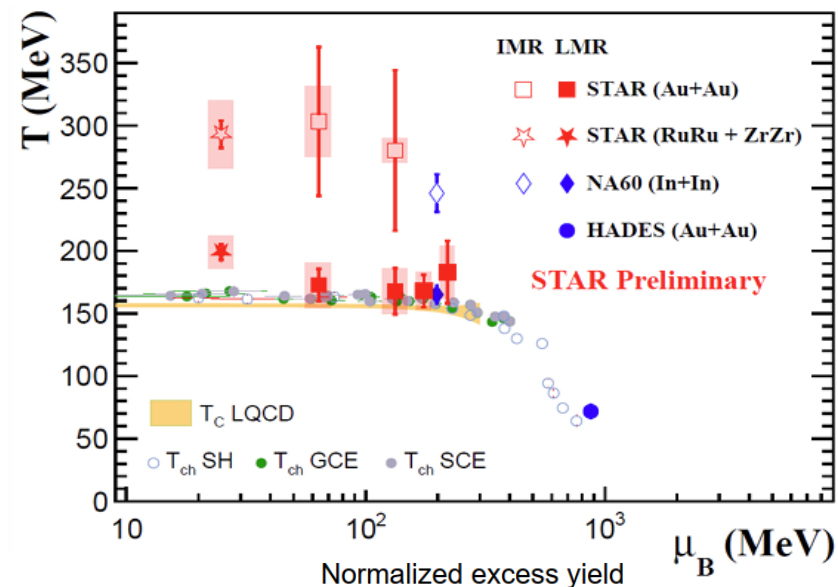
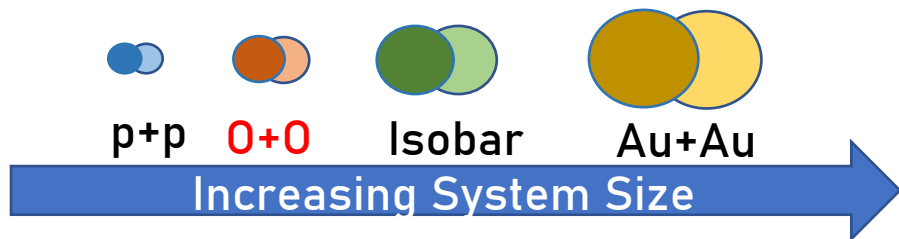
QGP: $M^{3/2} * e^{-M/T}$

In-med. ρ : Relativistic Breit-Wigner $* e^{-M/T}$



Nat. Phys. 15, 990-991 (2019)

- **Thermometer:** extract temperature from mass spectra
- **Chronometer:** predict lifetime from integrated yield.
- Thermal dileptons in **O+O**:
 - Do QGP exists in small systems?
 - What's the temperature/QGP lifetime in small system?



- Dataset: production_00_200GeV_2021, production_FF_00_200GeV_2021
- Year: 2021
- Production tag: P24ia
- Triggers used: minbias-bbc, minbias-epd (860001, 860002, 860011, 860012)
- Embedding request id: 20250301, 20250302, 20250303, 20250304

- **production_00_200GeV_2021:**

- 22130029, 22130030, 22130031, 22130046, 22130047, 22131011, 22131012, 22131013, 22131014, 22131015, 22131021, 22131033, 22131035, 22132014, 22132015, 22132024, 22132025, 22132026, 22133005, 22133007, 22133009, 22133018, 22133019, 22133020, 22133021, 22133022, 22133025, 22133031, 22133032, 22133037, 22133038, 22135030, 22135039

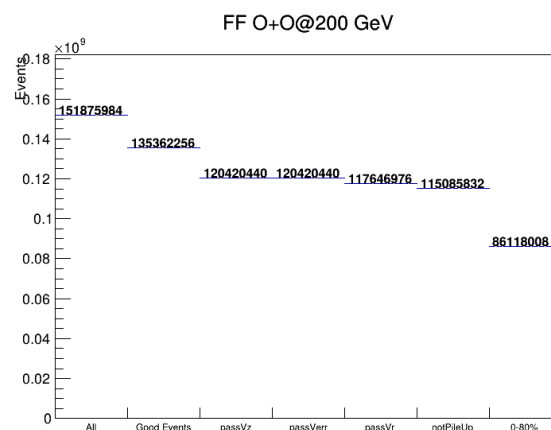
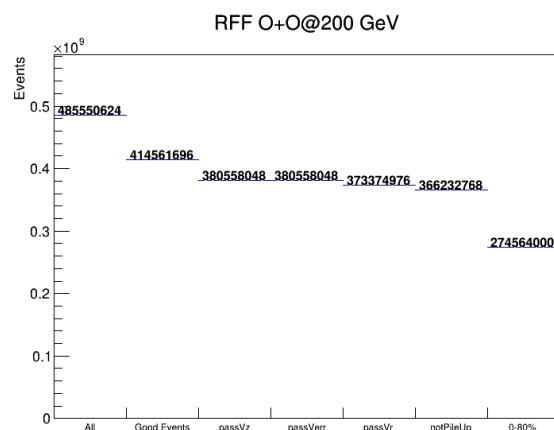
- **production_FF_00_200GeV_2021:**

- 22141038, 22141040, 22141043, 22142005, 22142077, 22143014, 22143015, 22143016, 22143029, 22143030, 22143031

Event level, Track level and PID cuts

- Event level:

- $|V_z| < 30$ cm
- $|V_r| < 2$ cm
- $V_x \geq 10^{-5}$ or $V_y \geq 10^{-5}$ or $V_z \geq 10^{-5}$ cm
- TPC-calibrated centrality (refmult6)
- ~421M weighted (~361M unweighted) events



- Details could be found in: <https://drupal.star.bnl.gov/STAR/meetings/STAR-Collaboration-Meeting-March-2025/LFSUPC-PWG-Parallel-Session-I/Raw-signal-dielectron-p>

- Track quality cut:

- TOF matched primary tracks
- $p_T \geq 0.2$ GeV/c
- $|\eta| < 1$
- nHitsFit ≥ 40
- nHitsFit/nHitsMax ≥ 0.52
- nHitsDedx ≥ 30
- DCA ≤ 1 cm
- $|Y_{Local}| < 1.8$

- PID cuts:

- $|1/\beta - 1| < 0.025$
- $p \leq 1$ GeV/c, $2.3p - 3.55 < n\sigma_e^{Corr} < 3$
- $p > 1$ GeV/c, $-1.25 < n\sigma_e^{Corr} < 3$

- Same ϕ_V cut as previous dielectron analysis:

- $\phi_V > 0.84e^{-49.5M_{ee}} - 0.997M_{ee} + 0.198$ for $0 < M_{ee} < 0.2$ GeV/c²

Analysis procedure: Raw Signal

- The invariant mass (M_{ee}) of the electron pairs:

$$M_{ee} = \sqrt{(E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2)^2}$$

- The Same-Event Like-Sign background:

$$N_{++\&--}^{corr} = 2\sqrt{N_{++} \cdot N_{--}} \times \frac{B_{+-}}{2\sqrt{B_{++} \cdot B_{--}}}$$

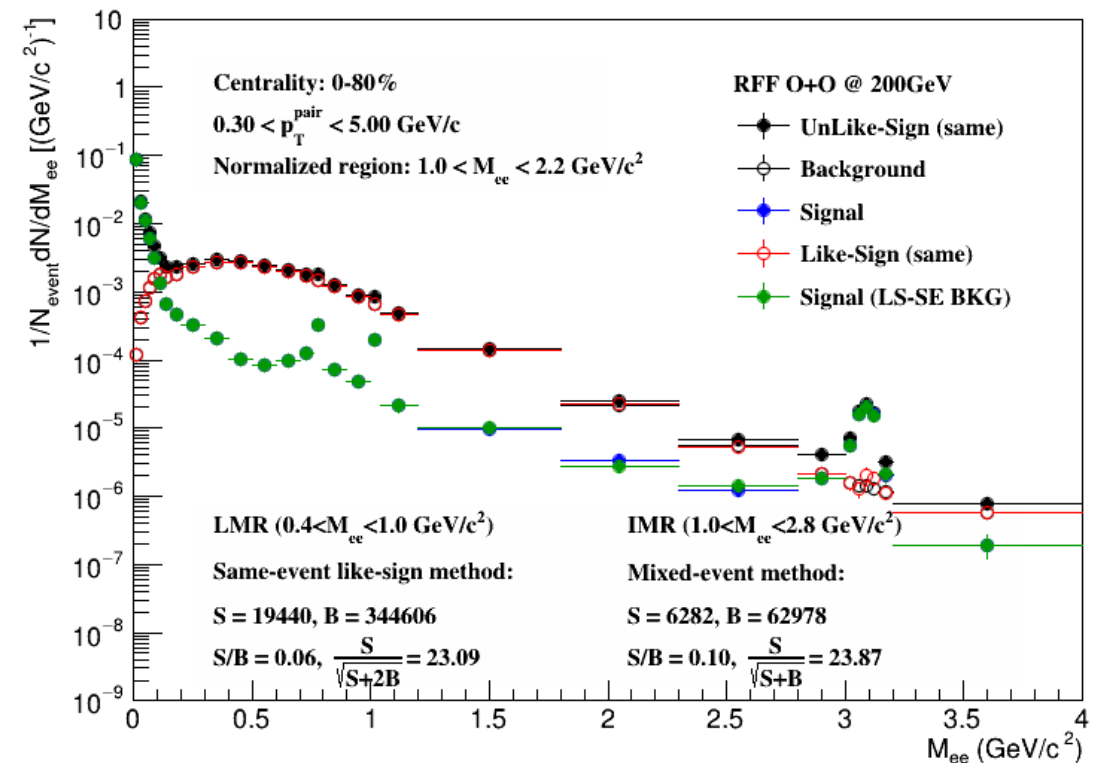
- The Mixed-Event Unlike-Sign background:

$$B_{Norm}^{+-} \times r(M_{ee}), \quad r(M_{ee}) = a + e^{\frac{M_{ee}-b}{c}}$$

- $r(M_{ee})$ fitted to LSSE/USME ratio.

$$N^{Raw} = N^{+-} - BKG = \begin{cases} N^{+-} - N_{afterPSAC}^{++\&--} & \text{for } M_{ee} < M_{th} \\ N^{+-} - B_{Norm}^{+-} \times r(M_{ee}) & \text{for } M_{ee} \geq M_{th} \end{cases}, M_{th} = 1.0 \text{ GeV}/c^2$$

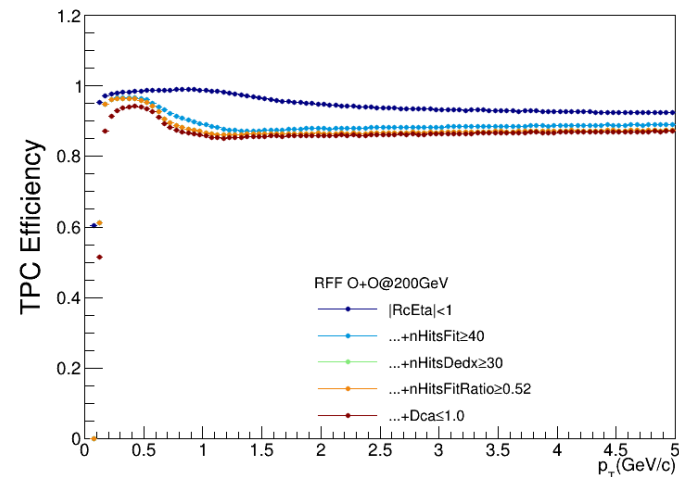
- Details could be found in: <https://drupal.star.bnl.gov/STAR/event/2026/01/26/LFSUPC-PWG-meeting/Update-dielectron-measurement-00-collisions-200-GeV>



Analysis procedure: Efficiency Correction

$$\epsilon_{\text{single}} = \epsilon_{\text{TPC}} \times \epsilon_{\text{TOFMatch}} \times \epsilon_{1/\beta} \times \epsilon_{n\sigma_e}$$

Centrality: 0-80%



TPC Tracking Efficiency

- From STAR official embedding simulation:

➤ TPC Tracking Efficiency

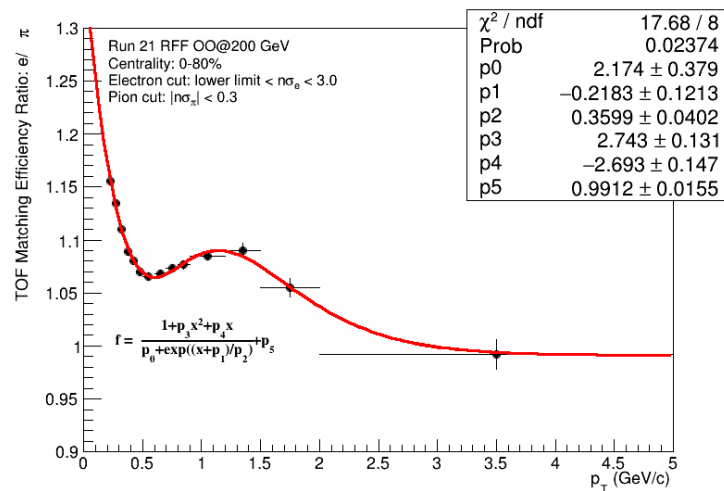
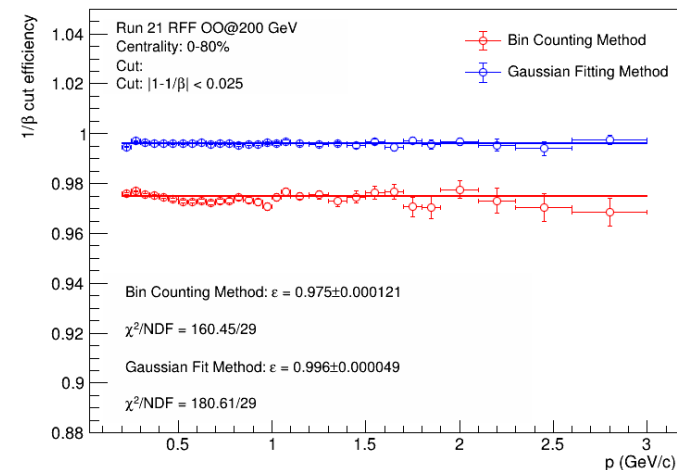
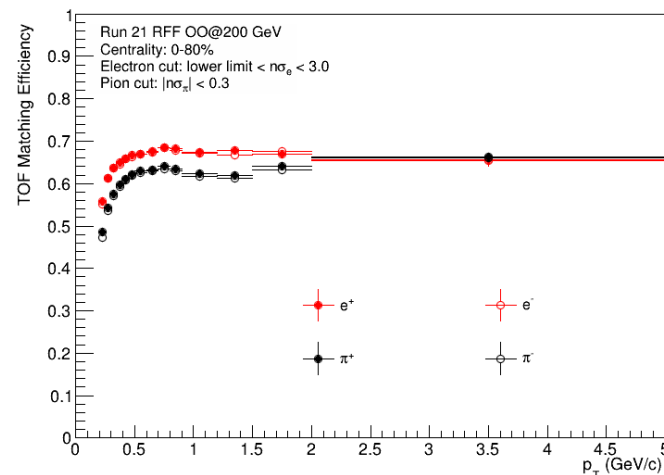
- From data-driven method:

➤ TOF Matching Efficiency

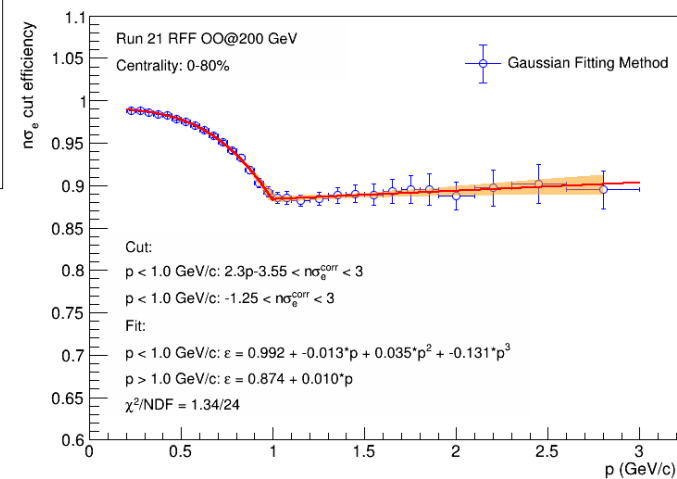
➤ PID Efficiency

▣ $1/\beta$ cut efficiency

▣ $n\sigma_e$ cut efficiency

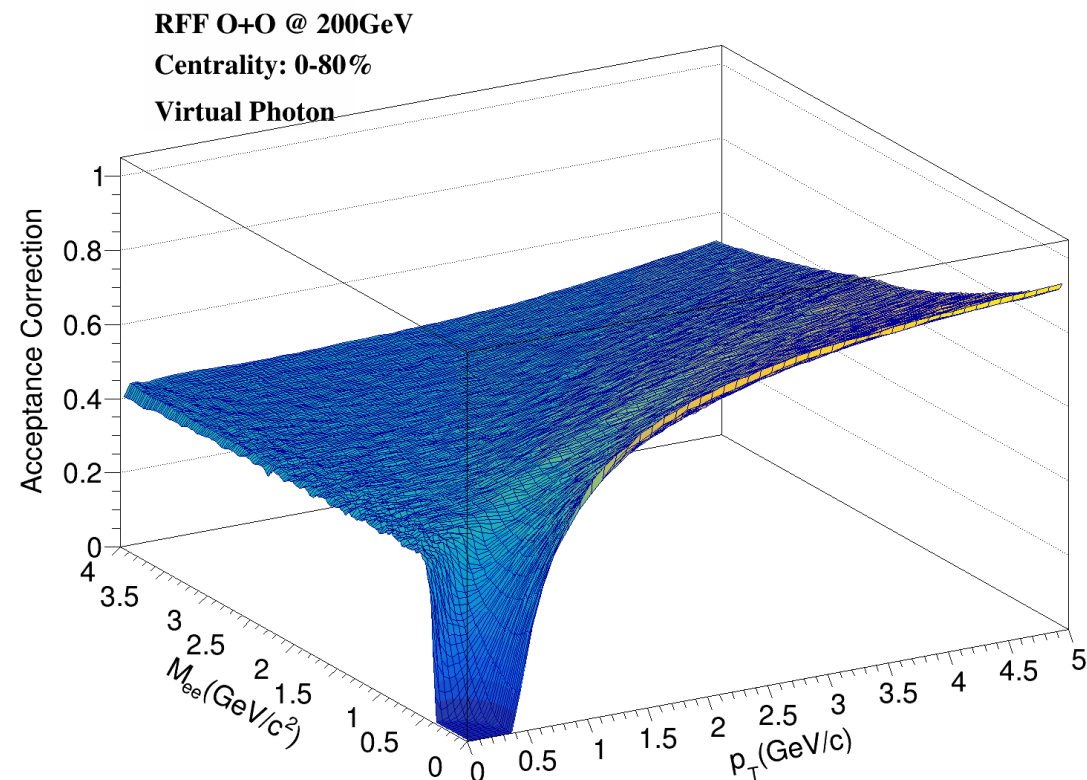
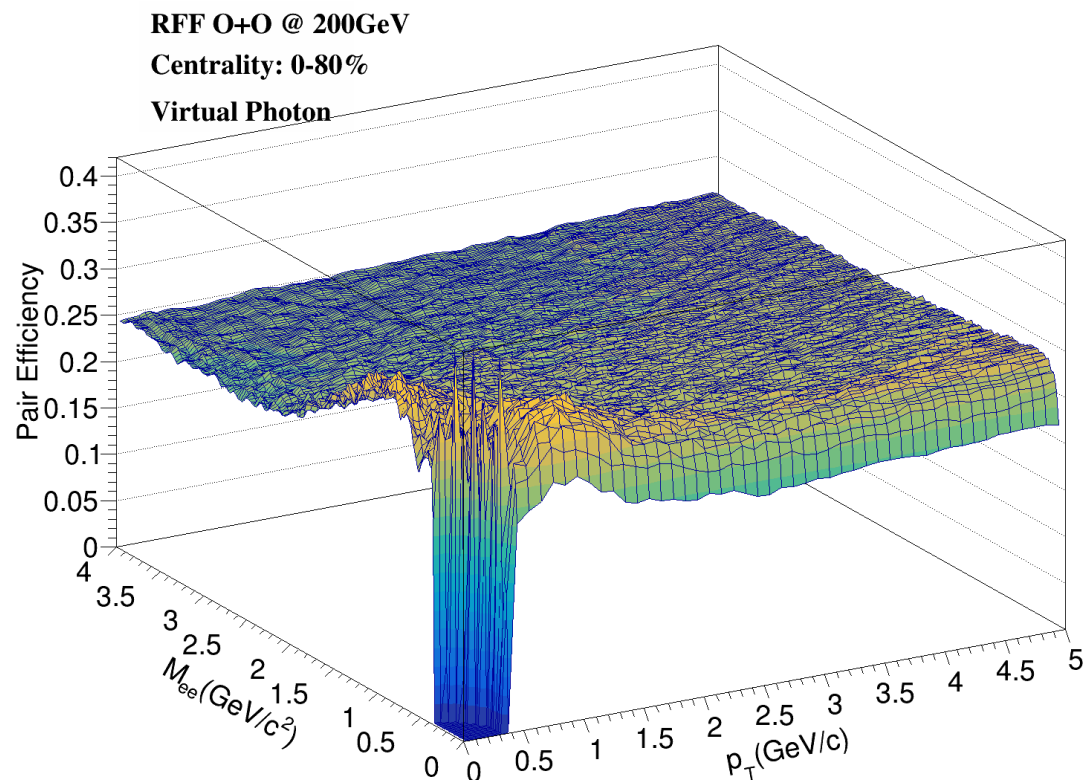


TOF Matching Efficiency



PID Efficiency

Analysis procedure: Pair Efficiency

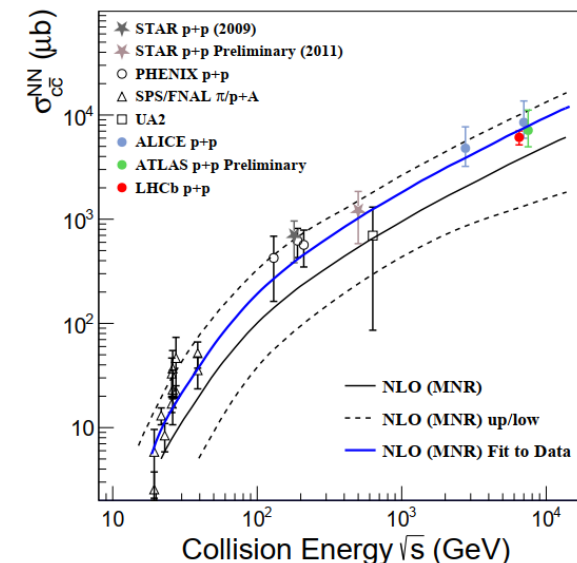


- Fold the single track efficiency within STAR acceptance
- Virtual photon as input.
- Calculated and applied in 2D (p_T and M_{ee}).

$$\varepsilon_{acc} = \frac{N_{pairs}(|y_{ee}| < 1, |\eta^e| < 1, p_T^e > 0.2 \text{ GeV/c})}{N_{pairs}(|y_{ee}| < 1)}$$

Analysis procedure: Hadronic Cocktail

1. Two-body decay: $\omega \rightarrow e^+e^-$, $\phi \rightarrow e^+e^-$, $J/\psi \rightarrow e^+e^-$
 2. Dalitz decay: $\pi^0 \rightarrow \gamma e^+e^-$, $\eta \rightarrow \gamma e^+e^-$, $\eta' \rightarrow \gamma e^+e^-$, $\omega \rightarrow \pi^0 e^+e^-$, $\phi \rightarrow \eta e^+e^-$
 3. Heavy-flavor decay: $c\bar{c}(\text{or } b\bar{b}) \rightarrow e^+e^- + X$
 4. Drell-Yan process: $q\bar{q} \rightarrow e^+e^-$
- Light hadrons: Toy MC
 - M_{ee} , rapidity input: follow formula in AuAu27/54.4 GeV analysis
 - p_T : Tsallis for ϕ & J/ψ ; TBW (60-70% isobar) for other particles.
 - Radiative decay are considered as well (e.g. $J/\psi \rightarrow \gamma ee$) in two-body decays.
 - HF/DY: Simulated in $p + p$ collisions using PYTHIA and then scale by N_{coll} .
 - PYTHIA6 Setting: follow the tuned parameters in 200GeV Au+Au and Isobar dielectron analysis
 - The NLO pQCD curve fitted to the world wide data for $c\bar{c}$: $\sigma_{c\bar{c}} = 443 \pm 28 \mu\text{b}$, $\sigma_{b\bar{b}} = 3.7 \mu\text{b}$, $\sigma_{DY} = 42 \text{ nb}$
 - Excess yield = Data - Cocktail



- Details could be found in: <https://drupal.star.bnl.gov/STAR/event/2026/01/26/LFSUPC-PWG-meeting/Update-dielectron-measurement-00-collisions-200-GeV>

Analysis procedure: Systematic uncertainty

- Background subtraction:
 - PSAC Factor: varying the mixed-event :
 - Pool Size: 100 $\rightarrow$ 200
 - $(V_z, \text{Centrality}, \Psi_2) = (20, 9, 12) \rightarrow (30, 16, 8)$
 - RFF/FF Difference for $M_{ee} < 0.2 \text{ GeV}/c^2$
- Efficiency correction:
 - TPC Tracking Efficiency:
 - DCA: 1.0 cm $\rightarrow$ 1.5 cm
 - (nHitsFit, nHitsDedx): (40, 30) $\rightarrow$ (30, 20)
 - TOF Matching Efficiency: $M_{ee} < 0.015 \text{ GeV}/c^2 \rightarrow M_{ee} < 0.01(0.02) \text{ GeV}/c^2$
 - $1/\beta$ Efficiency: difference between bin counting method and fitting method
 - $n\sigma_e$ Efficiency: $[n\sigma_e^{\text{Low}}, n\sigma_e^{\text{High}}] \rightarrow [n\sigma_e^{\text{Low}} \pm 0.2, n\sigma_e^{\text{High}} \pm 0.2]$
- Cocktail simulation:
 - Scaling factors: π^0 & ϕ yield, meson-to- π^0 ratio, decay BR, $\sigma_{c\bar{c}}$, $\sigma_{b\bar{b}}$, σ_{DY} , $\langle N_{coll} \rangle$
 - $c\bar{c} \rightarrow ee$: decorrelation (random ϕ) and R_{AA} weight p_T distribution.
- Temperature fit range:
 - LMR: $[0.2, 1.2] \rightarrow [0.2, 1.0] \text{ GeV}/c^2$
 - IMR: $[1.0, 2.8] \rightarrow [1.2, 2.8] \text{ GeV}/c^2$
- Details could be found in: <https://drupal.star.bnl.gov/STAR/meetings/STAR-Collaboration-Meeting-March-2026/LFSUPC-PWG-parallel-session-III/Update-thermal-dielec>

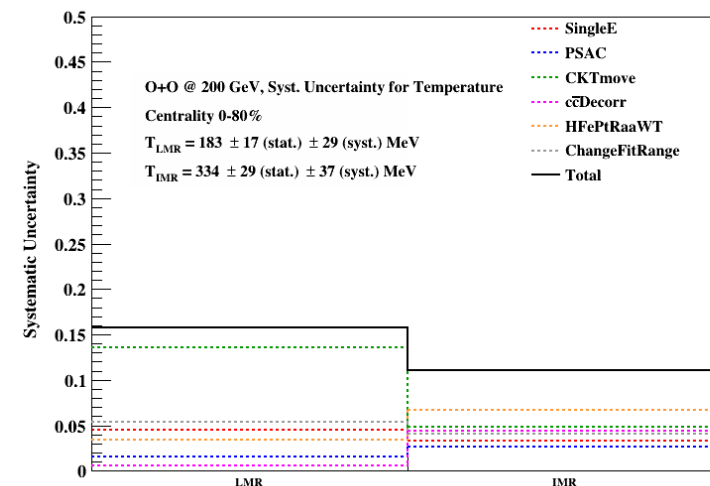
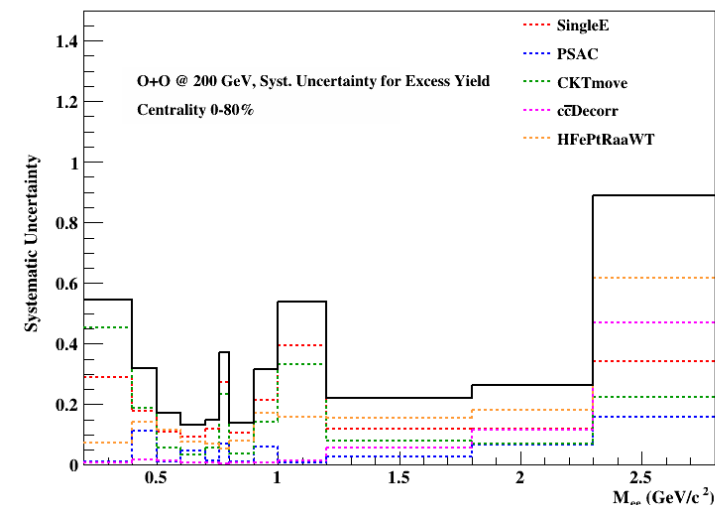
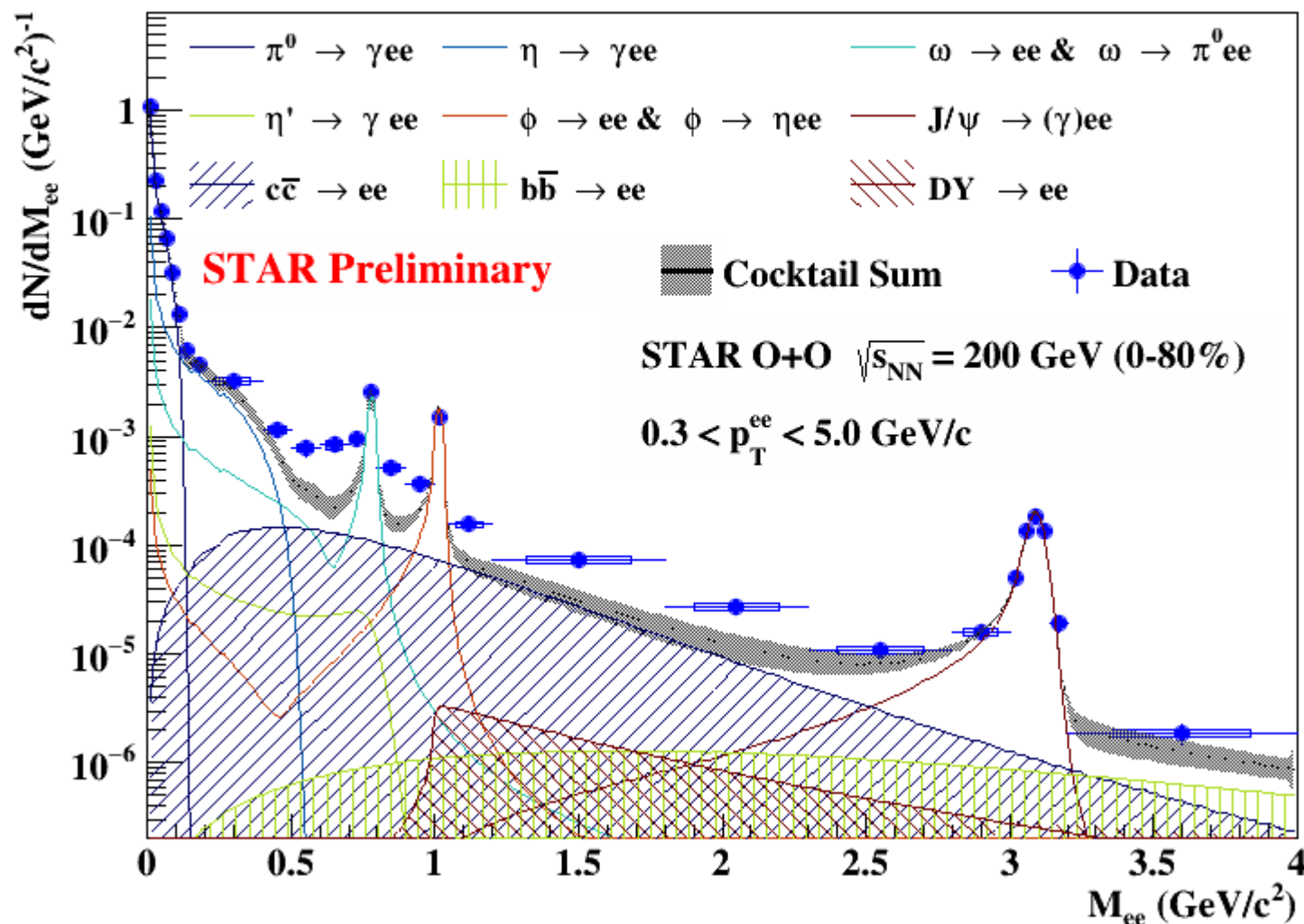


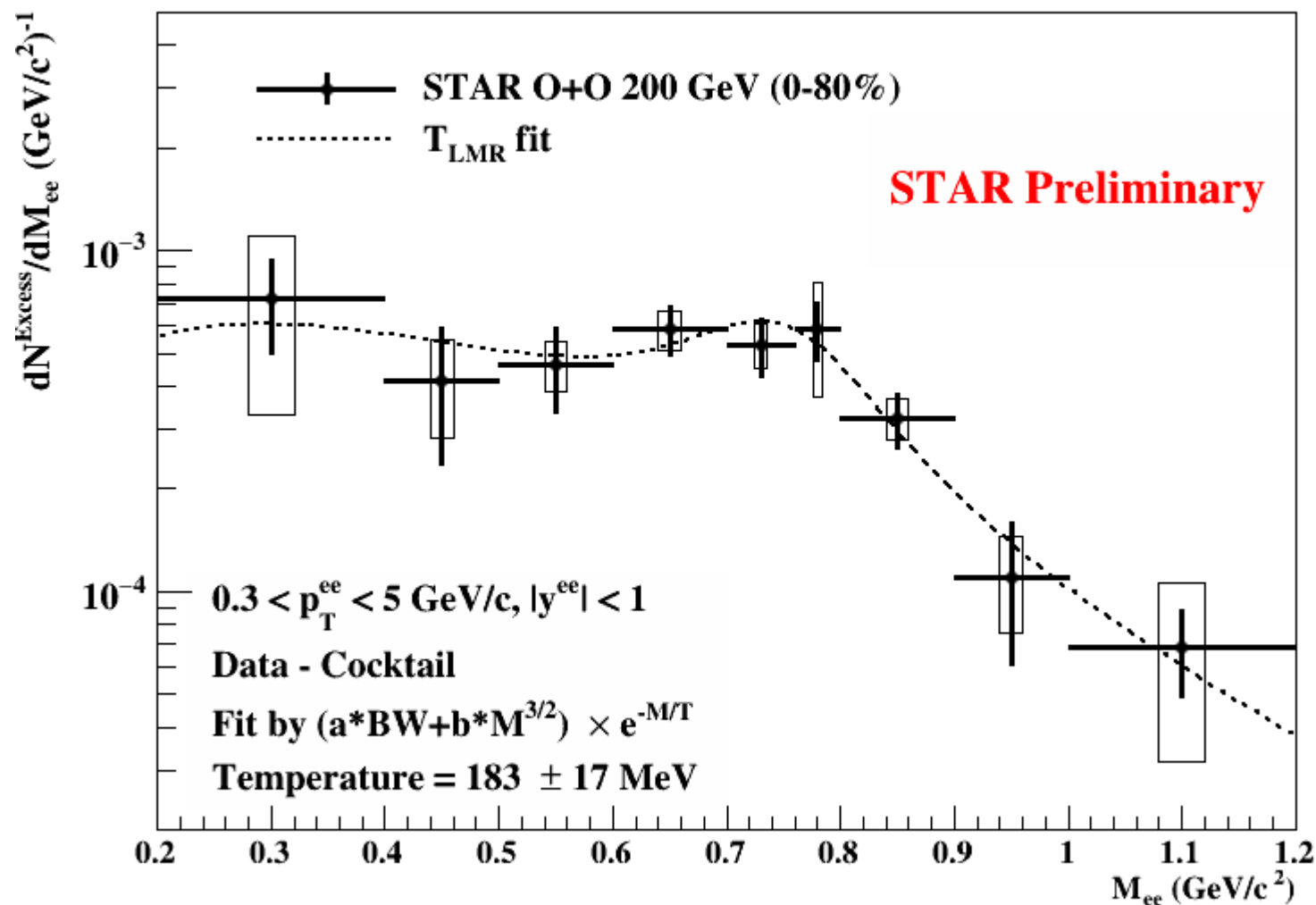
Figure 1



Physics message:

- Fully corrected data compared to cocktail.
- Data in very low mass region, well consistent with cocktail.
- Data in the low mass region (LMR) and intermediate mass region (IMR) show **clear enhancement** compared to the cocktail w/o ρ^0 contribution.

Figure 2

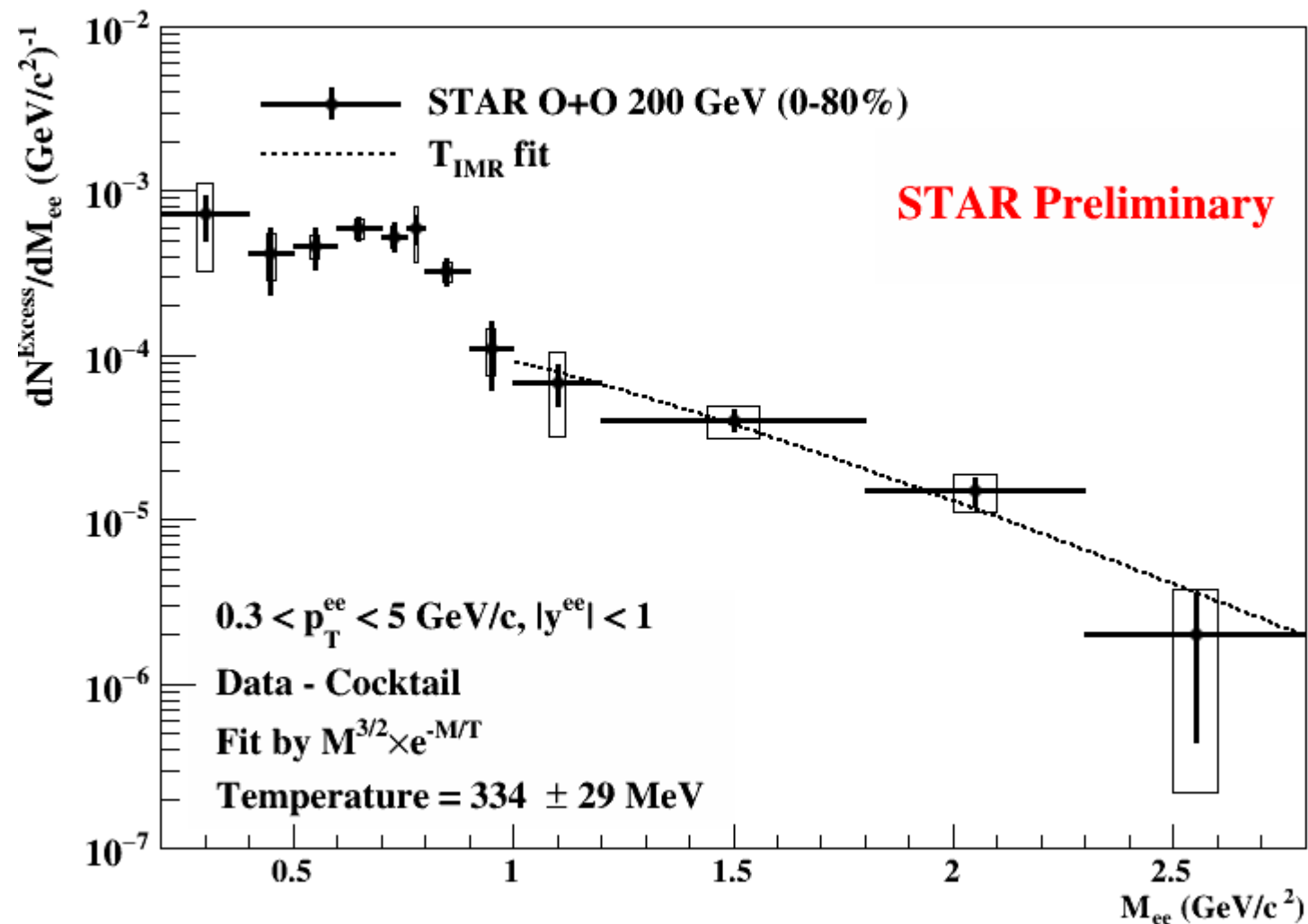


Physics message:

- Excess dielectron mass spectra in the low mass region (LMR).
- Fit function:

$$[a \times BW(M) + b \times M^{3/2}] \times e^{-M/T}$$
- Fit Range:

$$0.2 < M_{ee} < 1.2 \text{ GeV}/c^2$$
- $T_{\text{LMR}}^{\text{OO@200GeV}} = 183 \pm 17(\text{stat.}) \pm 29(\text{syst.}) \text{ MeV}$

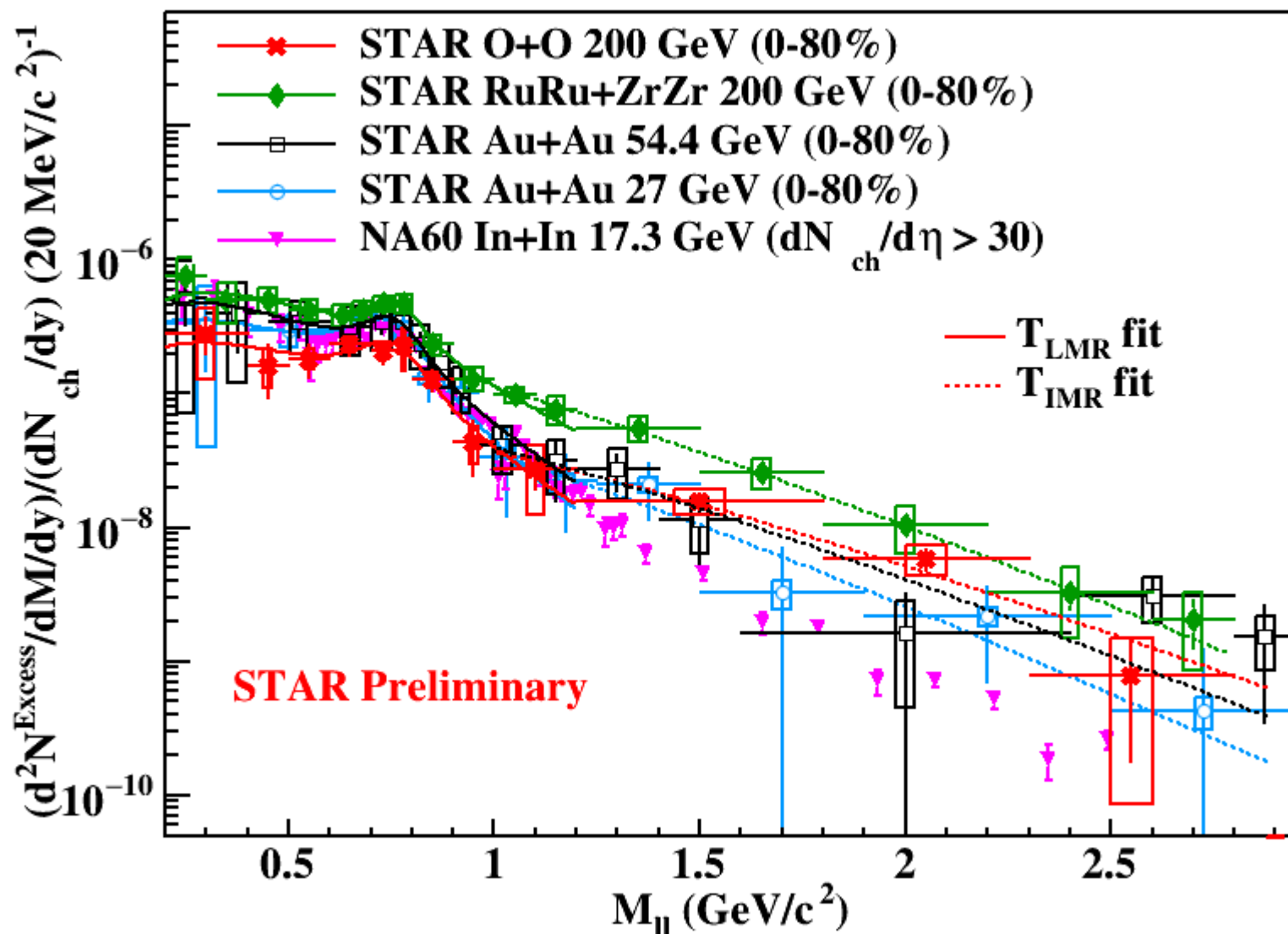


Physics message:

- Excess dielectron mass spectra in the intermediate mass region (IMR).
- Fit function:

$$a \times M^{3/2} \times e^{-M/T}$$
- Fit Range:

$$1.0 < M_{ee} < 2.8 \text{ GeV}/c^2$$
- $T_{\text{IMR}}^{\text{OO@200GeV}} = 334 \pm 29(\text{stat.}) \pm 37(\text{syst.}) \text{ MeV}$



Physics message:

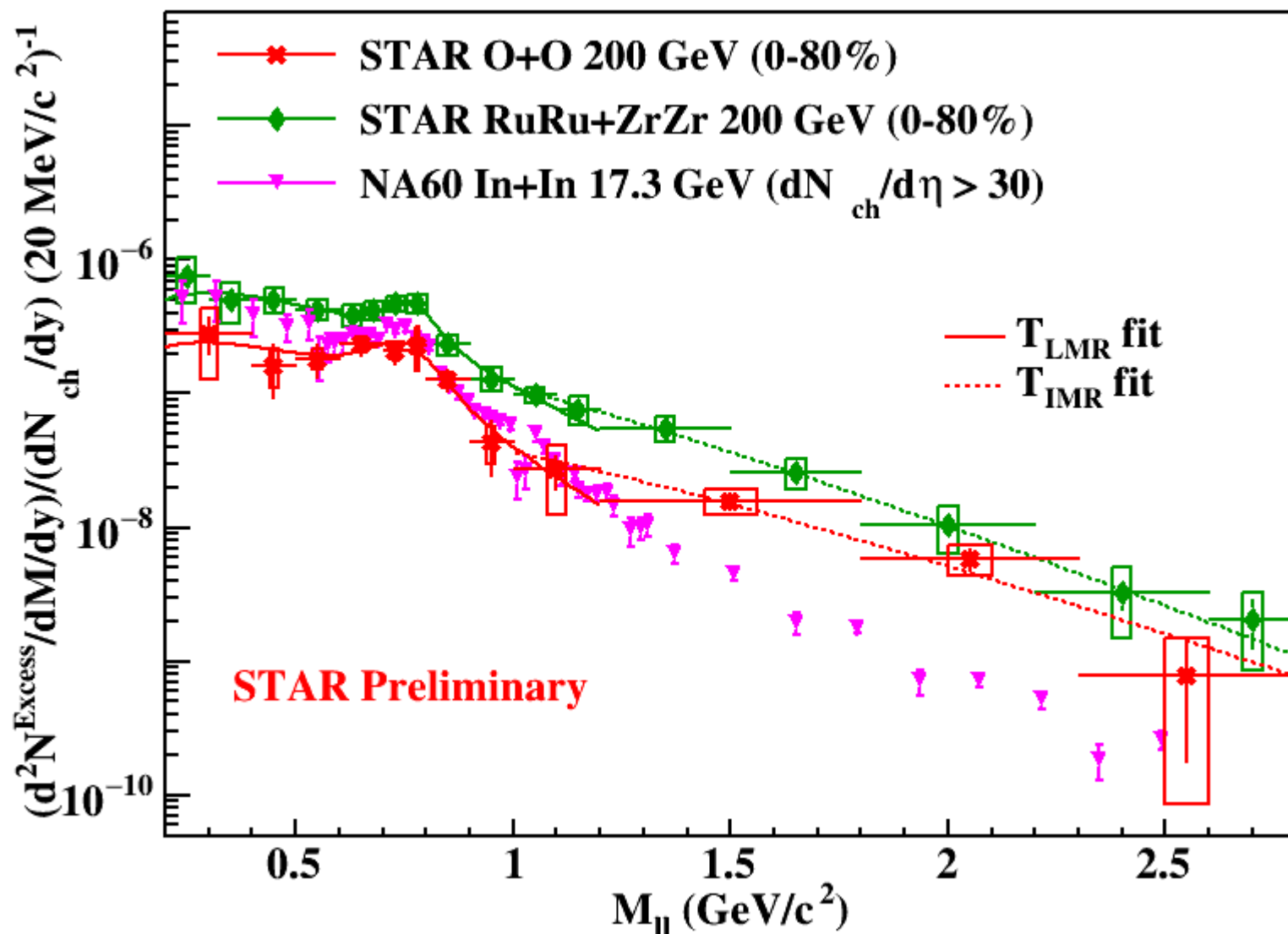
- Excess dielectron mass spectra in different collision systems.
- Update: **add O+O results**.
- Excess yield in 0+0 is smaller than those in larger collision systems in LMR.

NA60: EPJC 59, 607–623 (2009)

STAR 27 & 54.4 GeV: Nat Commun 16, 9098 (2025)

STAR isobar: Jiaxuan Luo's Ph.D. Thesis.

Figure 5



Physics message:

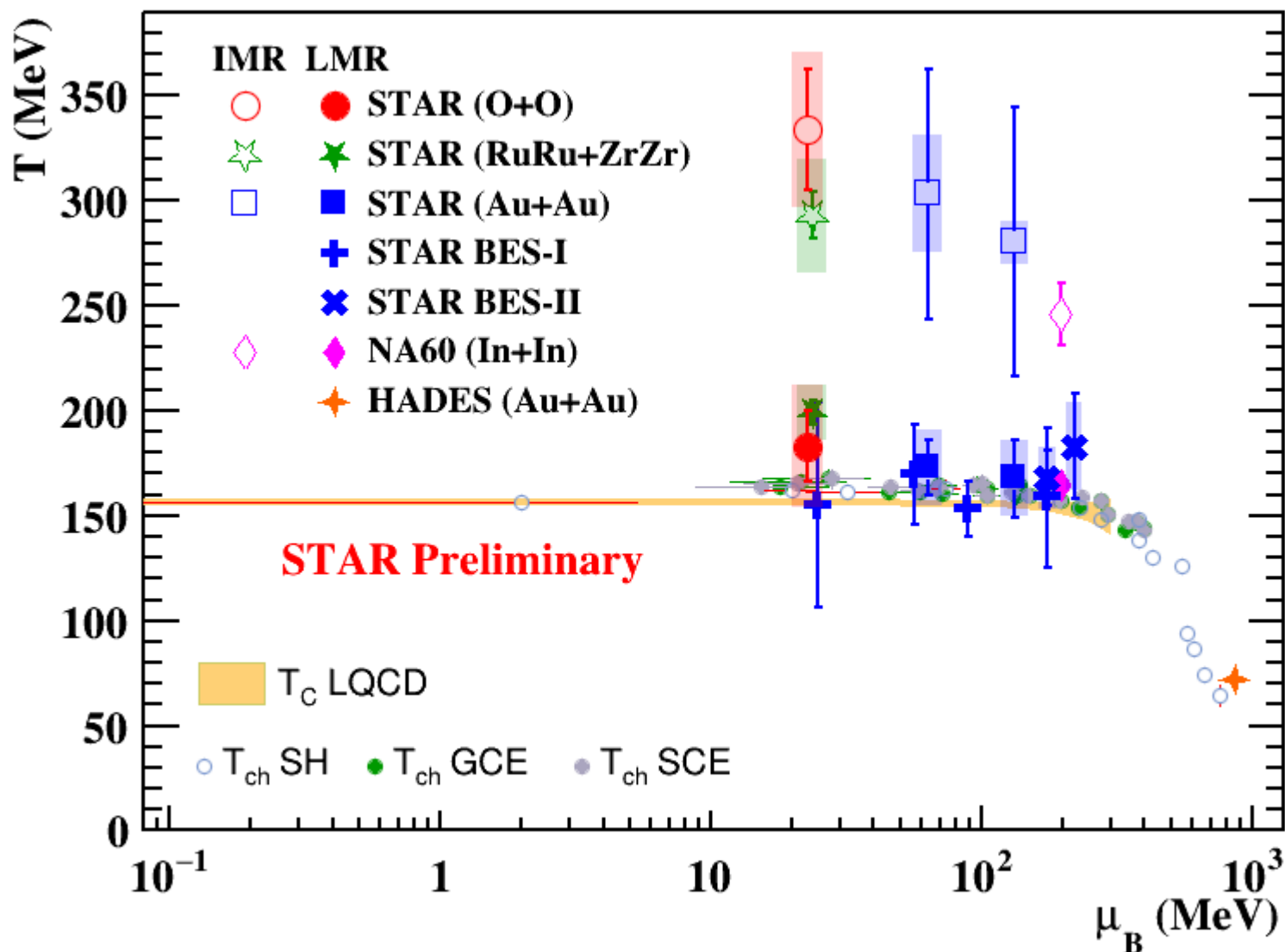
- Excess dielectron mass spectra in different collision systems.
- Update: **add O+O results**.
- LMR: Excess yield in O+O is smaller compared to NA60 and Isobar results.
- IMR: Excess yield in O+O follows similar trend as isobar's, while higher than NA60's results.

NA60: EPJC 59, 607–623 (2009)

STAR 27 & 54.4 GeV: Nat Commun 16, 9098 (2025)

STAR isobar: Jiaxuan Luo's Ph.D. Thesis.

Figure 6

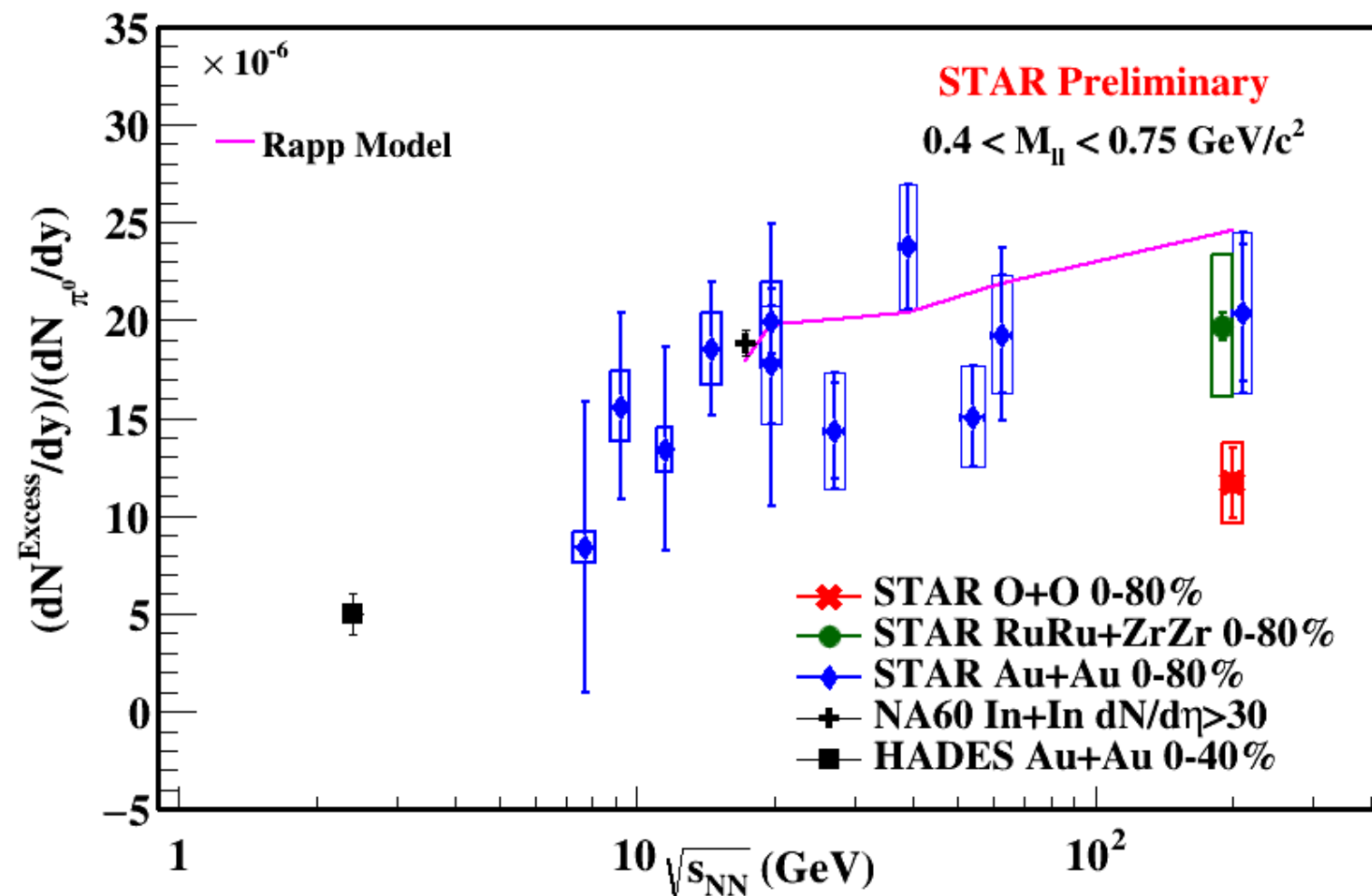


Physics message:

- Temperature vs. μ_B .
- Update: **add O+O results.**
- Both $T_{LMR}^{OO@200\text{GeV}}$ and $T_{IMR}^{OO@200\text{GeV}}$ are **consistent with isobar results** within uncertainties.
- Indicate QGP existence in 0+0 collisions.
- μ_B is just the estimation for comparison, not real.

NA60: EPJC 59,607–623(2009)
 HADES: Nat. Phys. 15, 1040–1045 (2019)
 HotQCD: Phys. Lett. B 795, 15–21 (2019)
 T_{ch} SH: P. Braun-Munzinger et al. Nat.561, 321–330 (2018)
 T_{ch} GCE/SCE: STAR Phys. Rev. C 96, 044904 (2017)
 STAR BES-I, 27 & 54.4 GeV: Nat Commun 16, 9098 (2025)
 STAR BES-II: Chenliang Jin's Ph.D. Thesis.
 STAR isobar: Jiaxuan Luo's Ph.D. Thesis.

Figure 7



Physics message:

- Excess yield normalized by the number of π^0 yield.
- Update: **add O+O results.**
- Normalized excess yield in 0+0 is smaller than those in larger systems @200GeV, indicating **smaller QGP lifetime** in small collision system.

STAR: Phys. Rev. C 107, L061901 (2023)

STAR: Phys. Lett. B 750, 64 (2015)

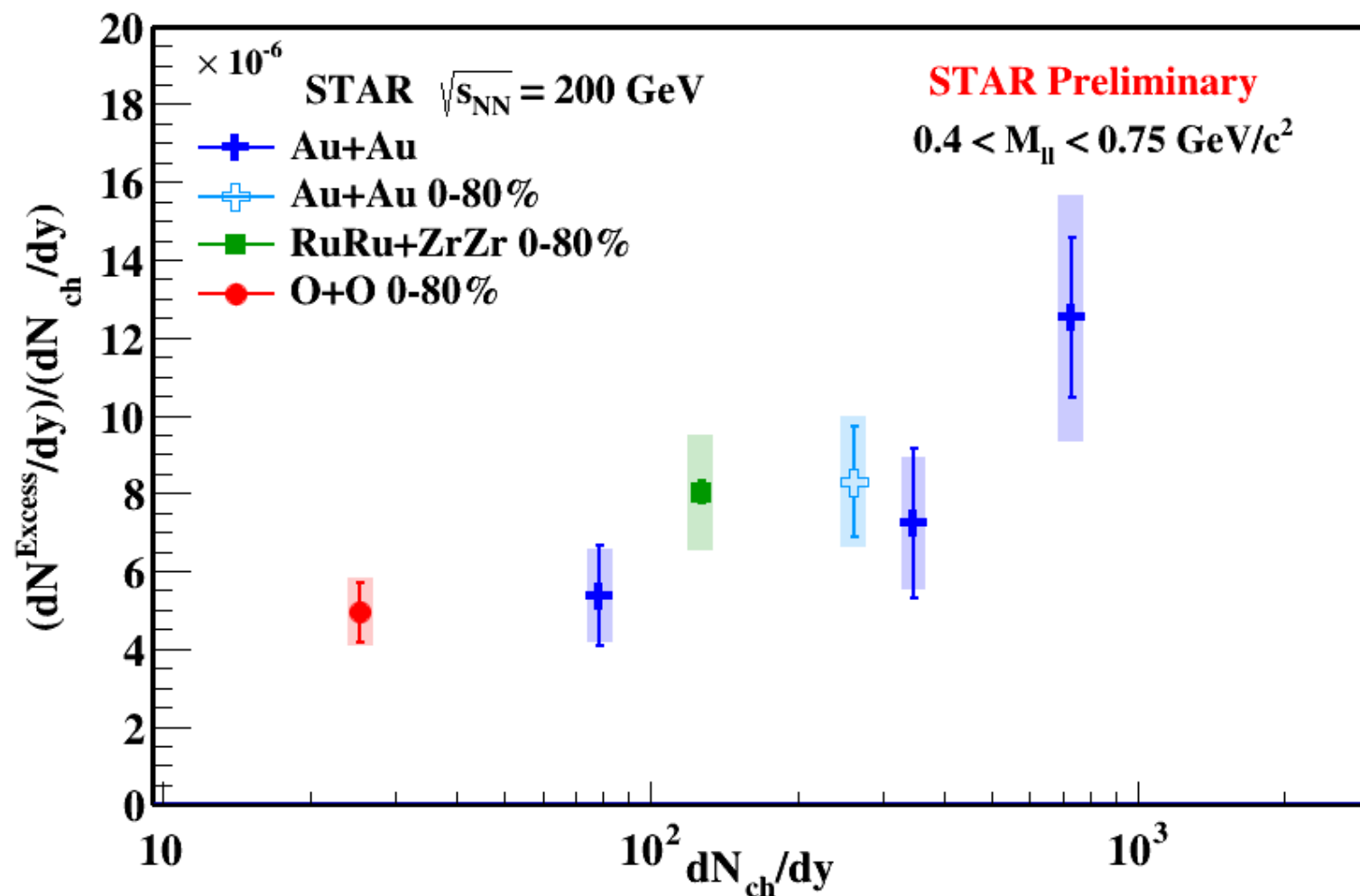
NA60: EPJC 59, 607–623 (2009)

HADES: Nat. Phys. 15, 1040–1045 (2019)

R. Rapp, Phys. Rev. C 63, 054907 (2001)

H. van Hees and R. Rapp, Phys. Rev. Lett. 97, 102301 (2006)

Figure 8



Physics message:

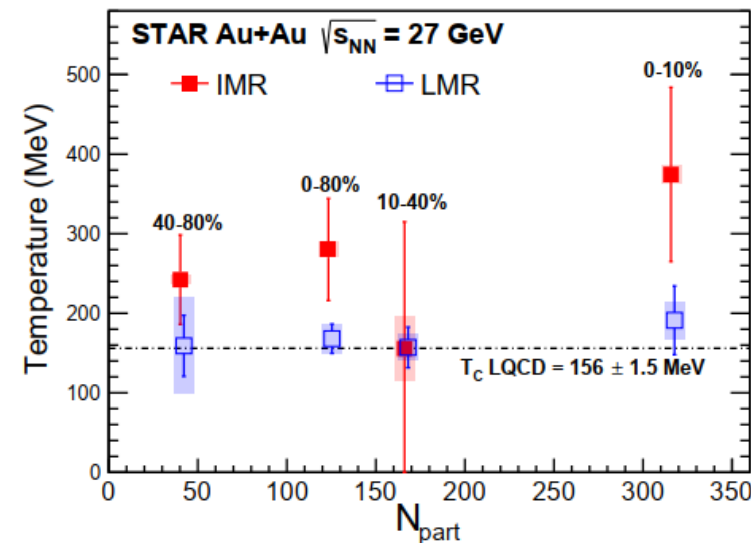
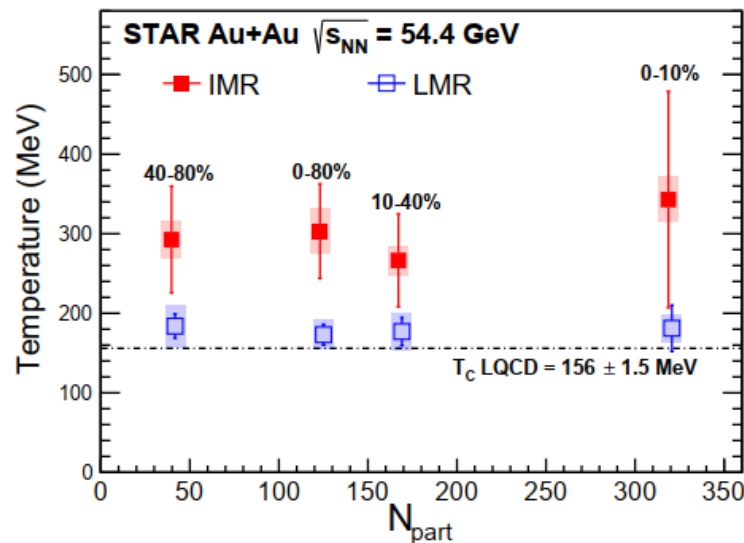
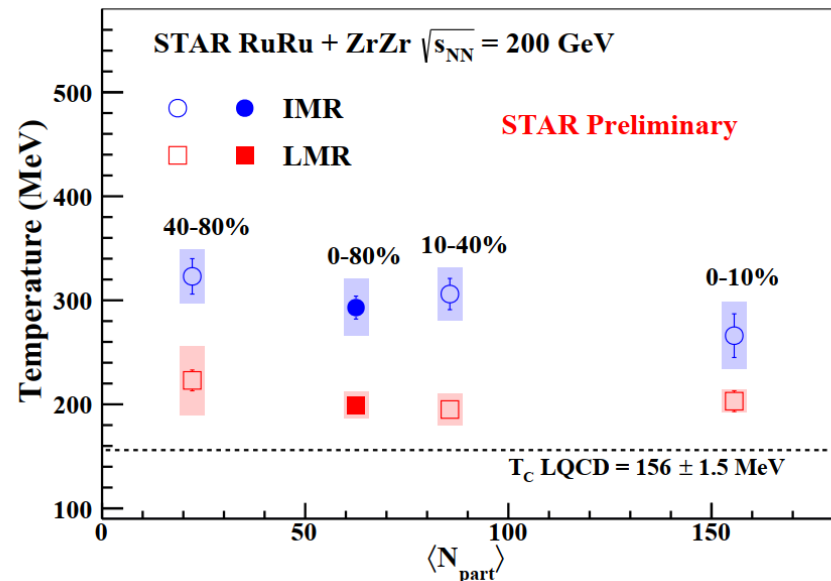
- Excess yield normalized by the number of charged particles.
- Normalized excess yield in 0+0 follows similar trend to the previous results.

STAR: Phys. Rev. C 107, L061901 (2023)
 STAR: Phys. Lett. B 750, 64 (2015)

Backup

Answers to comments from last talk

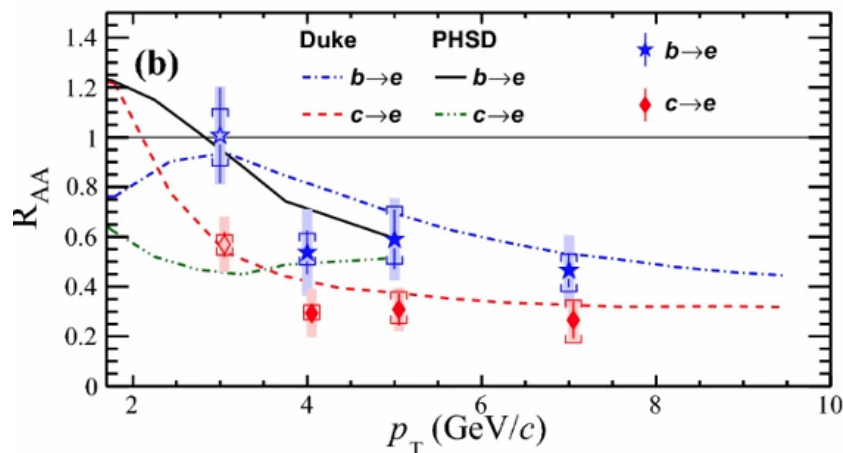
➤ What's the expected temperature? What physics message could be obtained from the results?



- No expected collision-system dependence for the extracted temperature.
 - No obvious $\langle N_{part} \rangle$ dependence for previous results.
 - The energy density is similar to that in isobar, only with smaller collision system. Therefore, the generated matter should have similar temperature. (from my understanding.)
- The results hint the possible existence of QPG in 0+0 collisions, with shorter lifetime compared to larger collision systems.

Answers to comments from last talk

- For $c\bar{c} \rightarrow ee$ process, considering the cold nuclear effect (CNM)?



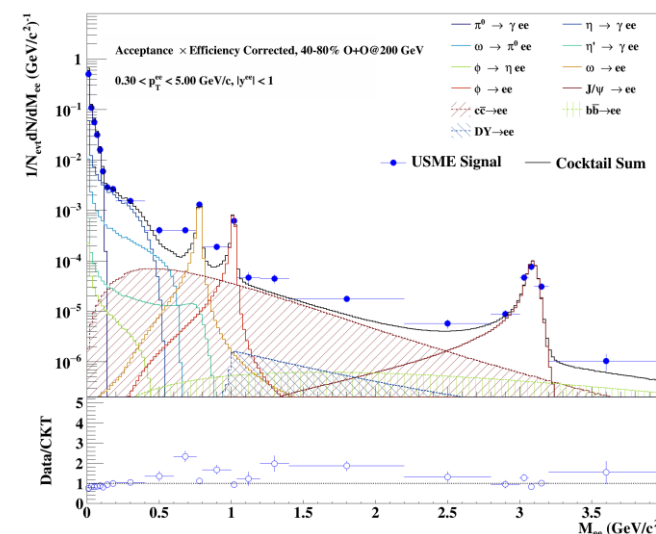
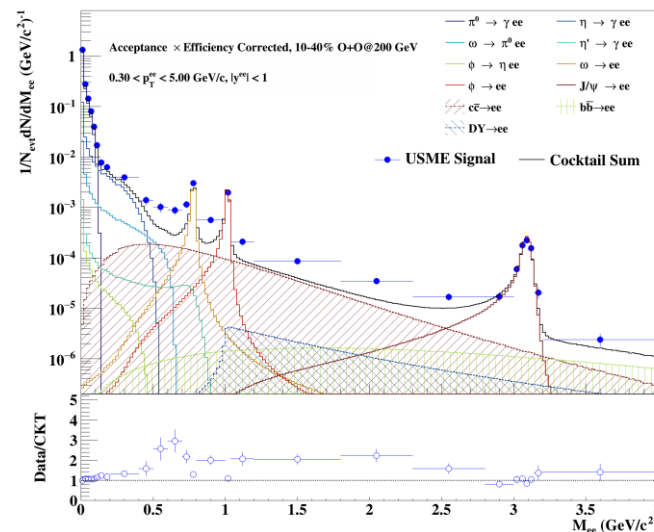
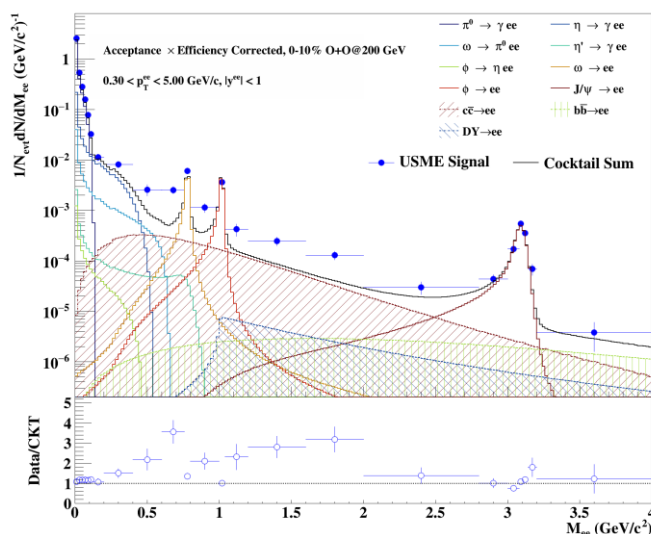
R_{AA} :STAR, Eur. Phys. J. C 82, 1150 (2022)
 Duke model: Phys. Rev. C 92, 024907 (2015)
 PHSD model: Phys. Rev. C 78, 034919 (2008)

- For the R_{AA} weight as the systematic uncertainty, the R_{AA} values are from the Duke and PHSD models fitted to the data in Au+Au collisions.
 - Such a method is also adopted in the isobar 200GeV analysis and the Au+Au 27/54.4 GeV analysis. Since the CNM effect would also play a role in Au+Au, such an R_{AA} weight p_T should have taken the CNM effect into consideration.
 - Besides, the CNM effect should little affect the results in 0+0 collisions due to very small $A=16$ for $^{16}_8\text{O}$, using the model fitted to Au+Au results to weight would only overestimate the uncertainties.
 - Moreover, we haven't found the effective way to calculate the effect according to nPDF now.

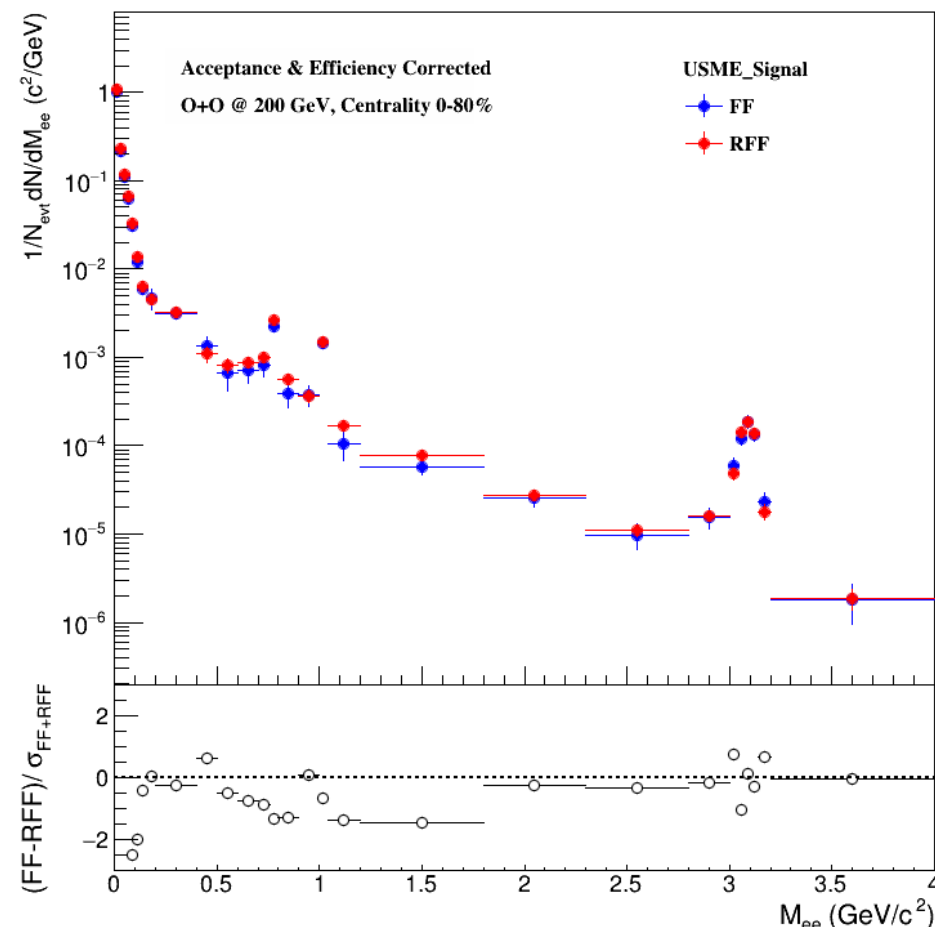
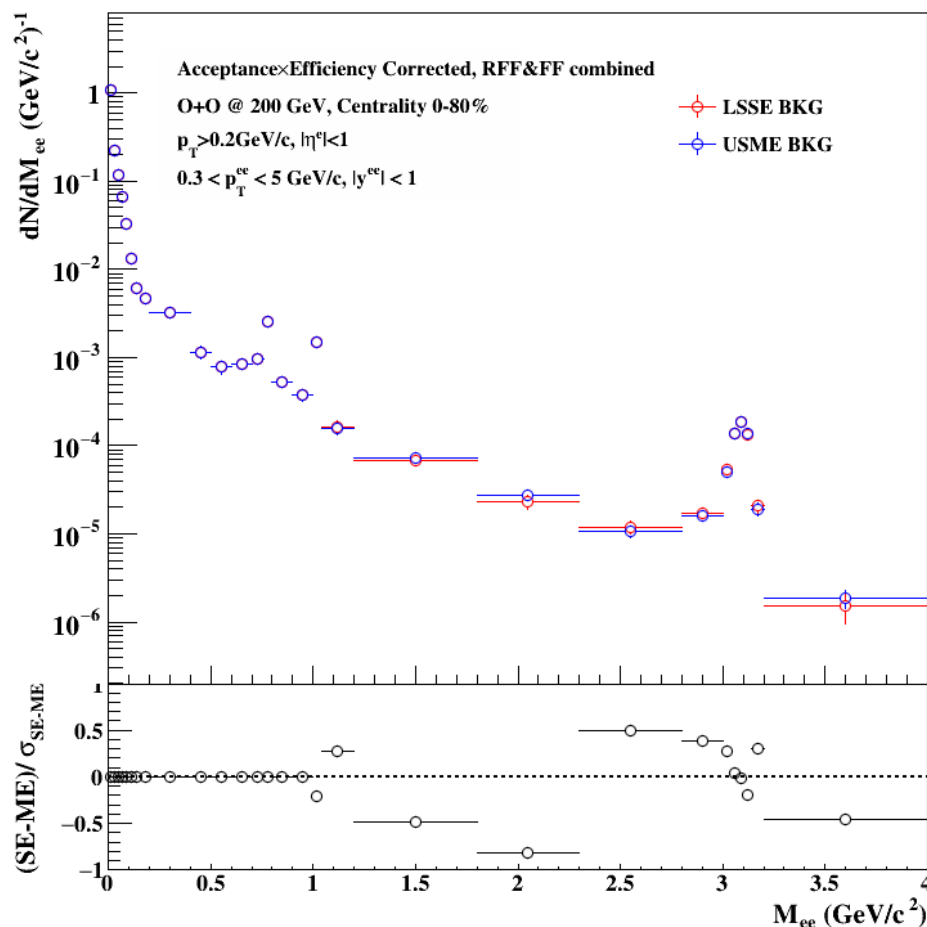
Answers to comments from last talk

➤ Results in different centrality bins?

Temporary results, not final ↓



- We did the possible centrality-differential results.
 - Since there's no much understanding about the nuclear shape and the nPDF about the Oxygen nuclei, the centrality in such a small collision system may not work very well, especially in peripheral collisions.
 - Besides, Aoke Zhang, who is focusing on the J/ψ analysis in O+O collisions, only requested preliminary results in 0-80% centrality bin. And the present cocktail relies on her results in $J/\psi \rightarrow (\gamma)ee$ process.
 - Therefore, we plan to request the results in 0-80% only, and other centrality results could be further studied in the future.



- The spectra with different bkg/field looks similar in most M_{ee} region, only add systematic uncertainty for FF/RFF in $M_{ee} < 0.2 \text{ GeV}/c^2$